

Glider

(Coach Scheduled Event - 45 Minutes)



Description: Teams will build and test one glider of any design using the cereal box they bring and the materials provided at the competition. They will also take a written assessment on principals of glider flight.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Safety Equipment - Teams Must Bring:

- **Safety glasses labeled ANSI Z87+ (impact rated)**
- **All competitors must wear their eye protection during the entire event.**
- **If a team does not have the required eye protection, they will be given the opportunity to obtain it, time allowing, but will not receive extra time.**
- **If a team is unable to obtain eye protection, the team will not compete and will receive a no-show score.**

Impact Safety Glasses Required



Eyeglasses are not safety rated!

Team members must also bring:

- 1-Family size cereal box - approximately 38 cm in height, 20 cm in width, and 7.6 cm in depth. These boxes usually contain between 18-20 ounces of cereal.
- Scissors
- Ruler or straight edge
- Writing utensils

Materials Provided at the Event:

- 1 meter of 1 inch-width masking tape
- 5 - Team choice Paperclips from the following:
 - Jumbo Paperclip 9 cm (3.5 in)
 - Standard size 25 mm (1 in) long
 - Teams may mix the sizes

THE COMPETITION

The Written Assessment: The written assessment will be given during the team's Construction Phase. The team members will answer questions on the principles of glider flight, to include but not limited to Lift, Drag, Weight, Thrust, Airfoil, Bernoulli's Principle, Center of Gravity (CG), Center of Lift (CL), Lift-to-Drag Ratio (L/D), Aspect Ratio, Dihedral, Reflex.

As a part of the assessment the team will estimate the time their glider will stay aloft.

Construction Phase (30 min)

- Teams will build their glider using the cereal box, and the materials provided. They do not have to use all of the materials, but must use at least two paperclips.
- Contestants may test and improve their devices at the discretion of the event supervisor, before their official launch.

- Teams must write their team number on their glider.

Glider Testing:

- Each team will have two official launches of their glider.
- All teams will release their glider from the same launch point marked on the floor of the launch area.
- Teams may not use a running start before they launch their glider. The team's feet must be on the floor when they launch their glider.
- Participants will announce their intention to launch their glider by saying "3, 2, 1, Launch."
- Time will start when judges see the glider leave the hand of the student.
- The Event Supervisors will record the time a team's glider stays airborne.
- Timing will continue until the glider touches the top of the launch area, any obstruction, or lands on the floor.
- The Event Supervisor will record all flight times and use the longest time as the official time score for the team. Times will be recorded in seconds to the best precision of the timing device.

SCORING: Final Score (FS) = Assessment Score + Time Score, max = 100 points

- Assessment Score = $(TAS \div CAS) \times 50$; rounded to the 100th place value; maximum 50 points
 - Team Assessment Score (TAS) = the team assessment score
 - Competition Assessment Score (CAS) = the highest assessment score for all teams
- Time Score = $(TT \div CT) \times 50$; rounded to the 100th place value; maximum 50 points
 - Competition Time (CT) = the longest flight timing of all teams
 - Team Time (TT) = the time the team's glider stays in the air

Scoring Example:

- Competition Assessment Score(CAS) = 15. The team scores 9 (TAS) on the assessment.
Assessment Score for the team= $(9 \div 15) \times 50 = 30$ pts
- Competition Time (CT) = 3.56 seconds. The team's time is 1.45 (TT).
Time Score for the team= $(1.45 \div 3.56) \times 50 = 20.36$ pts
- Final Score (FS) = $30 + 20.36 = 50.36$ points

Tiebreakers:

- 1st - The closest estimate to the actual flight time
- 2nd - Written Assessment Score

Resources:

- [How to Make Hand Gliders : 7 Steps - Instructables](#)
- [Cereal Box Glider w/ Canard and Delta Wing - YouTube](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles